30.3: The reproductive adaptations of angiosperms include flowers and fruits

On an evolutionary tree, a branch goes to gymnosperms and angiosperms; angiosperms are highlighted.

Branch point 1: First level-1 branch: Nonvascular plants (bryophytes). The second level-1 branch splits from branch point 2 into the first level-2 branch: seedless vascular plants. The second level-2 branch splits from branch point 3 into the first level-3 branch: Gymnosperms and the second level-3 branch: Angiosperms.

Commonly known as flowering plants, angiosperms are seed plants with the reproductive structures called flowers and fruits. The name *angiosperm* (from the Greek *angion*, container) refers to seeds contained in fruits. Angiosperms are the most diverse and widespread of all plants, with more than 290,000 species (about 90% of all plant species).

Characteristics of Angiosperms

All angiosperms are classified in a single phylum, Anthophyta. Before considering the evolution of angiosperms, we will examine two of their key adaptations—flowers and fruits—and the roles of these structures in the angiosperm life cycle.

Flowers

The **flower** is a unique angiosperm structure that is specialized for sexual reproduction. In many angiosperm species, insects or other animals transfer pollen from one flower to the sex organs on another flower, which makes pollination more directed than the wind-dependent pollination of most species of gymnosperms. However, some angiosperms *are* wind-pollinated, particularly those species that occur in dense populations, such as grasses and tree species in temperate forests.

A flower is a specialized shoot that can have up to four types of modified leaves called floral organs: sepals, petals, stamens, and carpels **(Figure 30.8)**. Starting at the base of the flower are the **sepals**, which are usually green and enclose the flower before it opens (think of a rosebud). Interior to the sepals are the **petals**, which are brightly colored in most flowers and can aid in attracting pollinators. Flowers that are wind-pollinated, such as grasses, generally lack brightly colored parts. In all angiosperms, the sepals and petals are sterile floral organs, meaning that they do not produce sperm or eggs.

Figure 30.8

The structure of an idealized flower.

The structure of an idealized flower.

At the top of the stem is the receptacle. On this is the ovary, in which are the ovules. Sepals extend to either side of the receptacle. Around the ovary are petals, and within this are stamen and single carpel (simple pistil). Stamen have a filament and an anther. Single carpel refers to the ovary, the style from the ovary, and the stigma at the end of the style.

Within the petals are two types of fertile floral organs that produce spores, the stamens and carpels. Stamens and carpels are sporophylls, modified leaves that are specialized for reproduction. **Stamens** are microsporophylls: They produce microspores that develop into pollen grains containing male gametophytes. A stamen consists of a stalk called the **filament** and a terminal sac, the **anther**, where pollen is produced. **Carpels** are megasporophylls: They produce megaspores that give rise to female gametophytes. The carpel is the "container" mentioned earlier in which seeds are enclosed; as such, it is a key structure that distinguishes angiosperms from gymnosperms. At the tip of the carpel is a sticky **stigma** that receives pollen. A **style** leads from the stigma to a structure at the base of the carpel, the **ovary**; the ovary contains one or more ovules. As in gymnosperms, each angiosperm ovule contains a female gametophyte. If fertilized, an ovule develops into a seed.

A flower may have one or more carpels. In many species, multiple carpels are fused into one structure. The term **pistil** is sometimes used to refer to a single carpel (a simple pistil) or two or more fused carpels (a compound pistil). Flowers also vary in symmetry **(Figure 30.9)** and other aspects of shape, as well as size, color, and odor. Much of this diversity results from adaptation to specific pollinators (see Figures 38.4 and 38.5).

Figure 30.9

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Flower symmetry.
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Compares radial symmetry (example is a daffodil) with bilateral symmetry (example is an orchid).

In radial symmetry, the sepals, petals, stamens, and carpels radiate out from a center. Any line through the central axis divides the flower into two equal parts. In bilateral symmetry, the flower can only be divided into two equal parts by a single line.

DRAW IT *Draw the single line that can divide the bilaterally symmetrical flower into two equal parts.*

*For suggested answer, see*Appendix A.**

Fruits

As seeds develop from ovules after fertilization, the ovary wall thickens and the ovary matures into a **fruit**. A pea pod is an example of a fruit, with seeds (mature ovules, the peas) encased in the ripened ovary (the pod).

Fruits protect seeds and aid in their dispersal. Mature fruits can be either fleshy or dry **(Figure 30.10)**. Tomatoes, plums, and grapes are examples of fleshy fruits, in which the wall (pericarp) of the ovary becomes soft during ripening. Dry fruits include beans, nuts, and grains. Some dry fruits split open at maturity to release seeds, whereas others remain closed. The dry, wind-dispersed fruits of grasses, harvested while on the plant, are major staple foods for humans. The cereal grains of maize, rice, wheat, and other grasses, though easily mistaken for seeds, are each actually a fruit with a dry outer covering (the former wall of the ovary) that adheres to the seed coat of the seed within.

Figure 30.10

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Some variations in fruit structure.
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Compares the different structures of the fruits tomato, ruby grapefruit, nectarine, hazelnut, and milkweed.

Part a Tomato, a fleshy fruit with soft outer and inner layers of pericarp (fruit wall).

Part b Ruby grapefruit, a fleshy fruit with a firm outer layer and soft inner layer of pericarp.

Part c Nectarine, a fleshy fruit with a soft outer layer and hard inner layer (pit) of pericarp.

Part d Hazelnut, a dry fruit that remains closed at maturity.

Part (e) Milkweed, a dry fruit that splits open at maturity.

As shown in **Figure 30.11**, various adaptations of fruits and seeds help to disperse seeds (see also Figure 38.12). The seeds of some flowering plants, such as dandelions and maples, are contained within fruits that function like parachutes or propellers, adaptations that enhance dispersal by wind. Some fruits, such as coconuts, are adapted to dispersal by water. And the seeds of many angiosperms are carried by animals. Some angiosperms have fruits modified as burrs that cling to animal fur (or the clothes of humans). Others produce edible fruits, which are usually nutritious, sweet tasting, and vividly colored, advertising their ripeness. When an animal eats the fruit, it digests the fruit's fleshy part, but the tough seeds usually pass unharmed through the animal's digestive tract. When the animal defecates, it may deposit the seeds, along with a supply of natural fertilizer, many kilometers from where the fruit was eaten.

Figure 30.11

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Fruit adaptations that enhance seed dispersal.
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Compares adaptations that promote seed dispersal: explosive action, wings, seeds within berries, and barbs.

Part a Some plants such as flowers have mechanisms that disperse seeds by explosive action which releases the seeds.

Part b Wings enable maple fruits to be carried by the wind.

Part c Seeds within berries and other edible fruits are often dispersed in animal feces after they are eaten.

Part d The barbs of cockleburs facilitate seed dispersal by allowing the fruits to hitchhike on animals such as a dog.

Animation

Fruit Structure and Seed Dispersal

image

Play

image

Play

The Angiosperm Life Cycle

You can follow a typical angiosperm life cycle in **Figure 30.12**. The flower of the sporophyte produces microspores that form male gametophytes and megaspores that form female gametophytes. The male gametophytes are in the pollen grains, which develop within microsporangia in the anthers. Each male gametophyte has two haploid cells: a *generative cell* that divides, forming two sperm, and a *tube cell* that produces a pollen tube. Each ovule, which develops in the ovary, contains a female gametophyte, also known as an **embryo sac**. The embryo sac consists of only a few cells, one of which is the egg.

Figure 30.12

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The life cycle of an angiosperm.
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Life cycle of an angiosperm from the time when the megasporocyte in each ovule divides by meiosis through fertilization and when a seed germinates.

The cycle is as follows for a mature flower on sporophyte plant (2 n) with a carpel and anther.

1. In the Megasporangium (2 n) of each ovule, the megasporocyte divides by meiosis, producing four megaspores (n). One survives and gives rise to a female gametophyte (embryo sac which includes the egg (n), synergids next to the egg, polar nuclei in the central cell, and antipodal cells.
2. On the anther, each microsporangium contains microsporocytes (2 n) that divide by meiosis, producing microspores (n).
3. A microspore develops into a pollen grain. The generative cell of the male gametophyte (n) will divide, forming two sperm. The tube cell (with tube nucleus) will produce the pollen tube.
4. After pollination (transfer to the stigma), two sperm cells are discharged, traveling down the pollen tube inside the style. Here, the two sperm are seen in the pollen tube, first behind the tube nucleus, and then eventually discharged into each ovule.
5. Double fertilization occurs. One sperm fertilizes the egg, forming a zygote. The other sperm fertilizes the central cell, forming the endosperm (a food supply, 3n in this example).
6. The zygote develops into an embryo that is packaged along with food into a seed. (Fruit tissues surround the seed).
7. When a seed germinates, the embryo develops into a mature sporophyte.

VISUAL SKILLS *Based on this figure, what is the maximum number of seeds this flower could produce? To produce that number of seeds, at least how many pollen grains would have to germinate?*

*For suggested answer, see*Appendix A*.*

Figure 30.12 Walkthrough

The life cycle of an angiosperm

Watch an explanation of this figure in the Figure 30.12 Walkthrough: The Life Cycle of an Angiosperm.

image

image

Animation

Angiosperm Life Cycle

image

Play

image

After its release from the anther, the pollen is carried to the sticky stigma at the tip of a carpel. Although some flowers self-pollinate, most have mechanisms that ensure **cross-pollination**, which in angiosperms is the transfer of pollen from an anther of a flower on one plant to the stigma of a flower on another plant of the same species. Cross-pollination enhances genetic variability. In some species, stamens and carpels of a single flower may mature at different times, or they may be so arranged that self-pollination is unlikely.

The pollen grain absorbs water and germinates after it adheres to the stigma of a carpel. The tube cell produces a pollen tube that grows down within the style of the carpel. After reaching the ovary, the pollen tube penetrates through the **micropyle**, a pore in the integuments of the ovule, and discharges two sperm cells into the female gametophyte (embryo sac). One sperm fertilizes the egg, forming a diploid zygote. The other sperm fuses with the two nuclei in the large central cell of the female gametophyte, producing a triploid cell (3n). This type of **double fertilization**, in which one fertilization event produces a zygote and the other produces a triploid cell, is unique to angiosperms.

After double fertilization, the ovule matures into a seed. The zygote develops into a sporophyte embryo with a rudimentary root and one or two seed leaves called **cotyledons**. The triploid central cell of the female gametophyte develops into **endosperm**, tissue rich in starch and other food reserves that nourish the developing embryo.

What is the function of double fertilization in angiosperms? One hypothesis is that double fertilization synchronizes the development of food storage in the seed with the development of the embryo. If a particular flower is not pollinated or sperm cells are not discharged into the embryo sac, fertilization does not occur, and neither endosperm nor embryo forms. So perhaps double fertilization is an adaptation that prevents flowering plants from squandering nutrients on infertile ovules.

Another type of double fertilization occurs in some gymnosperm species belonging to the phylum Gnetophyta. However, double fertilization in these species gives rise to two embryos rather than to an embryo and endosperm.

As you read earlier, the seed consists of the embryo, the endosperm, and a seed coat derived from the integuments. An ovary develops into a fruit as its ovules become seeds. After being dispersed, a seed may germinate if environmental conditions are favorable. The coat ruptures and the embryo emerges as a seedling, using food stored in the endosperm and cotyledons until it can produce its own food by photosynthesis.

Angiosperm Evolution

Charles Darwin once referred to the origin of angiosperms as an “abominable mystery.” He was particularly troubled by the relatively sudden and geographically widespread appearance of angiosperms in the fossil record (about 100 million years ago, based on fossils known to Darwin). Recent fossil evidence and phylogenetic analyses have led to progress in solving Darwin’s mystery, but we still do not fully understand how angiosperms arose from earlier seed plants.

Fossil Angiosperms

Angiosperms are now thought to have originated in the early Cretaceous period, about 140 million years ago. By the mid-Cretaceous (100 million years ago), angiosperms began to dominate some terrestrial ecosystems. Landscapes changed dramatically as conifers and other gymnosperms gave way to flowering plants in many parts of the world. The Cretaceous ended 66 million years ago with mass extinctions of dinosaurs and many other animal groups and with further increases in the diversity and importance of angiosperms.

What evidence suggests that angiosperms arose 140 million years ago? First, although pollen grains are common in rocks from the Jurassic period (201 to 145 million years ago), none of these pollen fossils have features characteristic of angiosperms, suggesting that angiosperms may have originated after the Jurassic. Indeed, the earliest fossils with distinctive angiosperm features are of 130-million-year-old pollen grains discovered in China, Israel, and England. Early fossils of larger flowering plant structures include those of *Archaefructus***(Figure 30.13)** and *Leefructus*, both of which were discovered in China in rocks that are 125 million years old. Overall, early angiosperm fossils indicate that the group arose and began to diversify over a 20- to 30-million-year period—a less sudden event than was suggested by the fossils known during Darwin’s lifetime.

Figure 30.13

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An early flowering plant.
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An early flowering plant.

Part a. Archaefructus sinensis, a 125-million-year- old fossil. This herbaceous species had simple flowers and bulbous structures that may have served as floats, suggesting it was aquatic. Recent phylogenetic analyses indicate that Archaefructus may belong to the water lily group. Scale: 5 centimeters.

Part b. Artist's reconstruction of Archaefructus sinensis shows tree-like structure with carpel at the tips, stamen along the branches, and possible floats below the stamen.

Can we infer traits of the angiosperm common ancestor from traits found in early fossil angiosperms? *Archaefructus*, for example, was herbaceous and had bulbous structures that may have served as floats,

suggesting it was aquatic. But investigating whether the angiosperm common ancestor was herbaceous and aquatic also requires examining fossils of other seed plants thought to have been closely related to angiosperms. All of those plants were woody, indicating that the common ancestor was probably woody and probably not aquatic. As we'll see, this conclusion has been supported by recent phylogenetic analyses.

Angiosperm Phylogeny

Molecular and morphological evidence suggests that extant gymnosperm lineages had diverged from the lineage leading to angiosperms by 305 million years ago. Note that this does not imply that angiosperms originated 305 million years ago, but that the most recent common ancestor of extant gymnosperms and angiosperms lived at that time. Indeed, angiosperms may be more closely related to several extinct lineages of woody seed plants than they are to gymnosperms. One such lineage is the Bennettitales, an extinct group with flowerlike structures that may have been pollinated by insects **(Figure 30.14a)**. However, the Bennettitales and other similar lineages of extinct woody seed plants did not have carpels or flowers and hence are not classified as angiosperms.

Figure 30.14

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Angiosperm evolutionary history.
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Diagram of a longitudinal section through Bennettitales, and an evolutionary tree.

Part a. A close relative of the angiosperms? This reconstruction shows a longitudinal section through the flowerlike structures found in the Bennettitales, an extinct group of seed plants hypothesized to be more closely related to extant angiosperms than to extant gymnosperms.

Part b. Angiosperm phylogeny. This tree represents a current hypothesis of angiosperm evolutionary relationships, based on morphological and molecular evidence. Angiosperms originated about 140 million years ago. The dotted line indicates the uncertain position of the Bennettitales, a possible sister group to extant angiosperms.

Branch point 1: (roughly 300 million years ago (million years ago)). Living gymnosperms.

Branch point 2 (roughly 225 million years ago). Bennettitales (dotted line).

Branch Point 3 (roughly 145 million years ago). Amborella

Branch Point 4 (roughly 140 million years ago). Water lilies.

Branch Point 5 (roughly 135 million years ago). Star anise and relatives.

Branch Point 6 (roughly 130 million years ago). Magnoliids.

Branch Point 7 (roughly 125 million years ago). Monocots. Eudicots.

VISUAL SKILLS *Would the branching order of the phylogeny in (b) necessarily have to be redrawn if a*150-million-year-old*fossil monocot were discovered? Explain.*

*For suggested answer, see**Appendix A*.

Making sense of the origin of angiosperms also depends on working out the order in which angiosperm clades diverged from one another. Here, dramatic progress has been made in recent years. Molecular and morphological evidence suggests that the shrub *Amborella trichopoda*, water lilies, and star anise are living representatives of lineages that diverged from other angiosperms early in the history of the group **(Figure 30.14b)**. *Amborella* is woody, supporting the conclusion mentioned earlier that the angiosperm common ancestor was probably woody. Like the Bennettitales, *Amborella*, water lilies, and star anise lack ***vessel elements***, efficient water-conducting cells that are found in most present-day angiosperms. Overall, based on the features of ancestral species and angiosperms like *Amborella*, researchers have hypothesized that early angiosperms were woody shrubs that had small flowers and relatively simple water-conducting cells.

Evolutionary Links with Animals

Plants and animals have interacted for hundreds of millions of years, and those interactions have led to evolutionary change. For example, herbivores can reduce a plant's reproductive success by eating its roots, leaves, or seeds. As a result, if an effective defense against herbivores originates in a group of plants, those plants may be favored by natural selection—as will herbivores that overcome this new defense. Plant-pollinator and other mutually beneficial interactions also can have such reciprocal evolutionary effects.

Plant-pollinator interactions also may have affected the rates at which new species form. Consider the impact of a flower's symmetry (see Figure 30.9). On a flower with bilateral symmetry, an insect pollinator can obtain nectar (a sugary solution secreted by flower glands) only when landing in a certain position **(Figure 30.15)**. This constraint makes it more likely that pollen is placed on a part of the insect's body that will come into contact with the stigma of a flower of the same species. Such specificity of pollen transfer reduces gene flow between diverging populations and could lead to increased rates of speciation in plants with bilateral symmetry. This hypothesis can be tested using the approach illustrated in this diagram:

Evolutionary tree showing the common ancestor which branches into Bilateral clade and Radial clade, branch length represents time since divergence from a common ancestor. Number of clade species can be compared.

Figure 30.15

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A bee pollinating a bilaterally symmetrical flower.
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A honeybee lands inside the petals of the Scottish broom flower and the stamens make contact with the back of the bee.

To harvest nectar from this Scottish broom flower, a honeybee must land as shown. This releases a tripping mechanism that arches the flower's stamens over the bee and dusts it with pollen. Later, some of this pollen may rub off onto the stigma of the next flower the bee visits.

Video

Bee Pollinating

image

image

A key step in this approach is to identify cases in which a clade with bilaterally symmetric flowers shares an immediate common ancestor with a clade whose members have radially symmetric flowers. One recent study identified 19 pairs of closely related "bilateral" and "radial" clades. On average, the clade with bilaterally symmetric flowers had nearly 2,400 more species than did the related clade with **radial symmetry**. This result suggests that flower shape can affect the rate at which new species form, perhaps by affecting the behavior of insect pollinators. Overall, plant-pollinator interactions may have contributed to the increasing dominance of flowering plants in the Cretaceous period, making angiosperms centrally important to ecological communities.

Angiosperm Diversity

From their humble beginnings in the Cretaceous period, angiosperms have diversified into more than 290,000 living species. Until the late 1990s, most systematists divided flowering plants into two groups, based partly on the number of cotyledons, or seed leaves, in the embryo. Species with one cotyledon were called **monocots**, and those with two were called **dicots**. Other features, such as flower and leaf structure, were also used to define the two groups. Recent DNA studies, however, indicate that the species traditionally called dicots are paraphyletic. The vast majority of species once categorized as dicots form a large clade, now known as **eudicots** ("true" dicots). **Figure 30.16** compares the main characteristics of monocots and eudicots. The rest of the former dicots are now grouped into four small lineages. Three of these lineages—*Amborella*, water lilies, and star anise and relatives—are informally called **basal angiosperms** because they diverged from other angiosperms early in the history of the group (see Figure 30.14b). A fourth lineage, the **magnoliids**, evolved later. **Figure 30.17** provides an overview of angiosperm diversity.

Figure 30.16

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Characteristics of monocots and eudicots.
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A table illustrates the differences in the following characteristics of monocots and dicots: embryos, leaf venation, stems, roots, pollen, and flowers.

blank	Embryos	Leaf venation	Stems	Roots	Pollen	Flowers
Monocot characteristics	One cotyledon	Veins usually parallel	Vascular tissue scattered	Root system usually fibrous (no main root)	Pollen grain with one opening	
Eudicot characteristics	Two cotyledons	Veins usually netlike	Vascular tissue usually arranged in ring	Taproot (main root) usually present	Pollen grain with three openings	Floral organs usually in multiples of four or five

Figure 30.17

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Exploring Angiosperm Diversity
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Exploring angiosperm diversity: photos and descriptions of basal angiosperms, magnoliids, monocots, and eudicots.

Descriptions are as follows.

Basal angiosperms: Surviving basal angiosperms consist of three lineages comprising only about 100 species. The first lineage to have diverged from other angiosperms is represented today by a single species, Amborella trichopoda (right). The other surviving lineages diverged later: a clade that includes water lilies and a clade consisting of the star anise and its relatives. Water lily (Nymphaea Rene Gerard). Species of water lilies are found in aquatic habitats throughout the world. Water lilies belong to a clade that diverged from other angiosperms early in the group's history. Amborella trichopoda. This small shrub, found only on

the South Pacific island of New Caledonia, may be the sole survivor of a branch at the base of the angiosperm tree. Star anise (Illicium). This genus belongs to a third surviving lineage of basal angiosperms.

Magnoliids: Magnoliids consist of about 8,000 species, most notably magnolias, laurels, and black pepper plants. They include both woody and herbaceous species. Although they share some traits with basal angiosperms, such as a typically spiral rather than whorled arrangement of floral organs, magnoliids are more closely related to eudicots and monocots. Southern magnolia (Magnolia grandiflora). This member of the magnolia family is a large tree. The variety of southern magnolia shown here, called Goliath, has flowers that measure up to about a foot across.

Monocots: About one-quarter of angiosperm species are monocots — about 70,000 species. Some of the largest groups are the orchids, grasses, and palms. Grasses include some of the most agriculturally important crops, such as maize, rice, and wheat. Examples pictured are orchid (Lemboglossum rossii); barley (Hordeum vulgare), a grass; and pygmy date palm (Phoenix roebelenii).

Eudicots: More than two-thirds of angiosperm species are eudicots — roughly 170,000 species. The largest group is the legume family, which includes such crops as peas and beans. Also important economically is the rose family, which includes many plants with ornamental flowers as well as some species with edible fruits, such as strawberry plants and apple and pear trees. Most of the familiar flowering trees are eudicots, such as oak, walnut, maple, willow, and birch. Examples pictured are snow pea (Pisum sativum), a legume; dog rose (Rosa canina), a wild rose; and pyrenean oak (Quercus pyrenaica).